TEAM T013 · CAUSAL NLP REASONING

Neuro-Symbolic Causal Inference

via Language Model Knowledge Extraction and Bayesian Networks

Divyansh Rathore
2401020021

Arya
2401020012

Krishna Dave
2401010238

Rishita Boisnobi
2401010382

Vriha Dholiya
2401010508

01 · PROBLEM

Language models read causal claims, but can’t reason with them

Goal: turn cause→effect claims in scientific text into a model that answers **“what happens if we intervene?”**

GAP 1

No guaranteed structure

Extracted claims can contradict each other: A→B and B→A.

GAP 2

No calibrated uncertainty

A confident model can still be wrong.

GAP 3

No intervention queries

“Seeing” X is not “doing” X. Text can’t compute P(Y | do(X)).

02 · APPROACH

Split the job: neural reading, symbolic reasoning



Every edge keeps its source abstract and sentence ID, so any arrow can be traced back and checked.

03 · DATA

Three datasets, each for one job

Dataset	Used for	Size
SemEval-2010 Task 8	Train and score both extractors (labelled)	10,698 sentences
PubMed abstracts	Build the causal graph (smoking & lung disease)	500 abstracts
NHANES 2013–2018	Probabilities for the Bayesian network	16,904 adults
Surgeon General reports	Reference facts to grade the graph	14 conclusions

Why not Corr2Cause? It tests logic over abstract variables (“A correlates with B”). There are no real entities to extract, so it can’t train an extractor.

04 · PREPARATION

Data preparation designed for a fair test

Labels → causal + direction

19 labels become 3 classes. Two of every three causal sentences name the effect first, so the arrow is flipped.

Imbalance: 12.5% causal

Accuracy would reward “never causal” with 88%. We report precision, recall and F1, and the direction must be right.

Leakage removed

19 sentences appeared in two splits. They were dropped from train/validation, and splits are made by document.

One held-out test set

2,717 official test sentences, used once at the end. Tuning used train and validation only.

Grammar rules over spaCy dependency parses

"The **burst** was caused by water hammer **pressure**."

burst = passive subject · pressure = agent
→ **pressure causes burst**

8 patterns: active, passive, participle, relative clause, "stems from", "cause of", "result of", bare "from". Grammar sets the arrow, not word order.

Trigger words pruned on train data

Validation	P	R	F1
24 intuitive verbs	0.56	0.63	0.59
9 kept (≥50% precise)	0.94	0.49	0.65

Dropped: "make" (7% precise), "produce" (25%), "from" (43%). These mostly signal other relations.

06 · DL MODEL

DistilBERT, fine-tuned with entity markers

The **<e1>**burst**</e1>** was caused by water hammer **<e2>**pressure**</e2>**.
→ **e2 causes e1** · e1 causes e2 · none

- Same 3-class task and inputs as the rules
- AdamW, lr 5×10^{-5}, batch 32, warm-up, 4 epochs
- Best epoch chosen on validation
- ~20 minutes on an Apple M3 GPU

Training curve

Epoch	Train loss	Val F1
1	0.320	0.83
2	0.079	0.87
3 (kept)	0.046	0.88
4	0.015	0.88

Loss keeps falling, validation plateaus: the start of overfitting.

From 500 abstracts to a 19-edge causal graph

500 PubMed abstracts

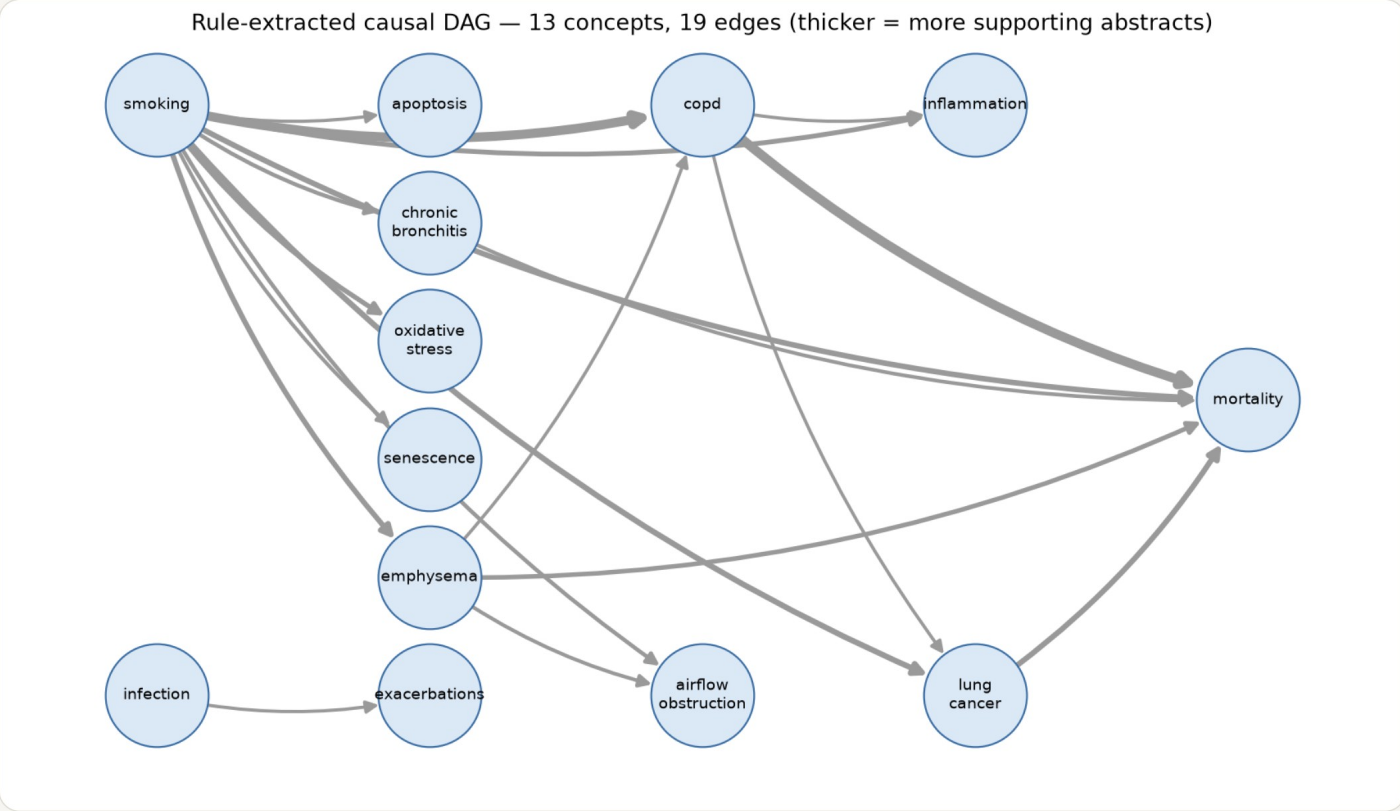
20 concepts, aliases merged

6,051 candidate pairs

54 edges extracted

20 in ≥ 2 abstracts

19 **after breaking cycles**



Text gives the arrows; survey data gives the numbers

- **Data:** 16,904 NHANES adults, linked to deaths up to 2019
- **Structure:** 11 text arrows + age → every node (a confounder)
- **Fit:** pgmpy, BDeu prior for rare cases (54 lung cancers)
- **Check:** 2 of 4 implied independences rejected. Reported, not patched.

Is the graph supported by data?

Structure	BIC ↑
No arrows	−46,728
Text arrows only	−45,740
Age arrows only	−45,276
Text + age	−45,058

The extracted arrows add information beyond age.

09 · SEEING VS DOING

Raw data says quitting smoking is dangerous

SEEING · RAW SURVEY

9.3% former smokers died

5.5% current smokers died

Former smokers are older. Age, not quitting, drives the gap.

DOING · P(DEATH | DO(SMOKING))

6.4% if all were former

6.9% if all were current

Same age mix for everyone: the order flips.

Six documented do-queries

#	Question	Answer	Matches evidence
Q1	Everyone smokes vs. nobody: how much more COPD?	9.3×	Yes
Q2	Share of COPD that wouldn’t exist without smoking	72%	Size not checked
Q3	Does quitting help? COPD, former vs. current	5.2% vs 10.4%	Expected
Q4	Smoking → lung cancer risk ratio	6.6×	Yes
Q5	Smoking → death during follow-up	1.8×	Yes
Q6	COPD → death during follow-up	1.7×	Yes

“Yes” = the direction matches a US Surgeon General conclusion (2014). The sources state that effects exist, not their exact size.

11 · FAIR COMPARISON

Each extractor wins somewhere; neither is enough

Evaluation	Rules (AML)	DistilBERT (DL)
SemEval test F1 [95% CI]	0.74 [0.69–0.78]	0.91 [0.88–0.93]
PubMed precision	0.79	0.45
PubMed recall	0.15	0.42
PubMed F1	0.25	0.43
Graph edge precision	0.58	0.22
Edges contradicting evidence	0	2
Established facts found (of 14)	11	13

PubMed: 300 silver labels from an instruction model, which was first tested on 150 human-labelled sentences (96% agreement, κ = 0.93).

12 · KEY INSIGHT

One flipped arrow changes the causal answer

smoking → COPD

9.3×

more COPD if everyone smoked

COPD → smoking

1.0×

“smoking has no effect”

DistilBERT proposed the reversed arrow in 33 abstracts, and it lost by only 4. Both graphs fit the survey data equally well; only the extracted structure decides.

What we would do with more time

Limitations

- PubMed labels are silver: not hand-checked
- Reference list covers 14 facts
- Self-reported diagnoses; short death follow-up
- Only age modelled as a confounder

Next steps

- Fine-tune on PubMed pairs, split by abstract
- Train on several pairs per sentence
- Human review of every extracted edge
- Add income, job exposure, spirometry

Fluent reading is not causal reasoning

1. The DL model reads far better on familiar text (F1 0.91 vs 0.74), but over-generates on scientific text.
2. The symbolic stage removes contradictions, tests the graph against data and answers do-queries.
3. Structure decides the answer: final queries agree with the published evidence.

Thank you · Questions?

Notebook, report and data: GitHub · 111-DEBUG-111
Reproducible: pinned versions, clean-run tested

APPENDIX

References

- Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems.* Morgan Kaufmann.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Devlin, J. et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. NAACL.
- Sanh, V. et al. (2019). DistilBERT, a distilled version of BERT. arXiv:1910.01108.
- Hendrickx, I. et al. (2010). SemEval-2010 Task 8: Multi-way classification of semantic relations between pairs of nominals.
- Jin, Z. et al. (2024). Can large language models infer causation from correlation? (Corr2Cause). ICLR.
- Ankan, A. & Panda, A. (2015). pgmpy: Probabilistic graphical models using Python. SciPy.
- U.S. DHHS (2014). *The Health Consequences of Smoking: 50 Years of Progress.* Report of the Surgeon General.
- CDC / NCHS. National Health and Nutrition Examination Survey 2013–2018, with linked mortality files.